

Tutorial 6: CS 215

October 30, 2025

1. Prove that the test statistic in a chi-square goodness of fit test has approximately a chi-square distribution.
Solution: Suppose we have k mutually exclusive categories with true probabilities $p_1, \dots, p_k > 0$, $\sum_{i=1}^k p_i = 1$.

From a sample of size n , let N_i be the number of observations in category i , so that

$$(N_1, \dots, N_k) \sim \text{Multinomial}(n; p_1, \dots, p_k).$$

The Pearson chi-square statistic is

$$X^2 = \sum_{i=1}^k \frac{(N_i - np_i)^2}{np_i}.$$

We show that under H_0 (i.e. the probabilities p_i are correct) and as $n \rightarrow \infty$,

$$X^2 \xrightarrow{d} \chi_{k-1}^2.$$

Let

$$\bar{N} = \frac{1}{\sqrt{n}}(N_1 - np_1, \dots, N_k - np_k)^\top.$$

The covariance matrix of N is

$$\text{Cov}(N) = n\Sigma, \quad \Sigma = \text{diag}(p_1, \dots, p_k) - pp^\top,$$

where $p = (p_1, \dots, p_k)^\top$. By the multivariate central limit theorem,

$$\bar{N} \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Now we write X^2 as a quadratic form. Let $D = \text{diag}(p_1, \dots, p_k)$. Then

$$X^2 = \sum_{i=1}^k \frac{(N_i - np_i)^2}{np_i} = \bar{N}^\top D^{-1} \bar{N}.$$

Thus asymptotically X^2 is a quadratic form in a mean-zero normal vector with covariance Σ .

Set $Y = D^{-1/2} \bar{N}$. Then $X^2 = Y^\top Y = \|Y\|^2$. The limiting distribution of Y is multivariate normal with mean 0 and covariance

$$\text{Cov}(Y) = D^{-1/2} \Sigma D^{-1/2} = I - uu^\top, \quad u = (\sqrt{p_1}, \dots, \sqrt{p_k})^\top.$$

Note that $I - uu^\top$ is a symmetric idempotent matrix (a projection) of rank $k - 1$.

Because $\text{Cov}(Y) = I - uu^\top$ is a projection onto the $(k - 1)$ -dimensional subspace orthogonal to u , there exists an orthonormal matrix Q such that

$$Q^\top (I - uu^\top) Q = \begin{pmatrix} I_{k-1} & 0 \\ 0 & 0 \end{pmatrix}.$$

Let $W \sim \mathcal{N}(0, I_k)$ be a standard k -variate normal vector. Then $(I - uu^\top)^{1/2}W$ has covariance $I - uu^\top$, so the limiting distribution of Y equals that of $(I - uu^\top)^{1/2}W$. Writing $Z = Q^\top(I - uu^\top)^{1/2}W$, we have

$$Z \sim \mathcal{N}\left(0, \begin{pmatrix} I_{k-1} & 0 \\ 0 & 0 \end{pmatrix}\right),$$

so $Z = (Z_1, \dots, Z_{k-1}, 0)^\top$ with $Z_1, \dots, Z_{k-1}$ i.i.d. $\mathcal{N}(0, 1)$. Hence

$$\|Y\|^2 \xrightarrow{d} \sum_{i=1}^{k-1} Z_i^2 \sim \chi_{k-1}^2.$$

Since $X^2 = \|Y\|^2$, we obtain the desired result.

If the probabilities p_i depend on s parameters estimated from the data, then s additional constraints are imposed and the effective rank of the projection is reduced by s . Consequently,

$$X^2 \xrightarrow{d} \chi_{k-1-s}^2.$$

Note: This is an asymptotic result: the approximation is good when all expected counts np_i are reasonably large (typically $np_i \geq 5$).

2. Prove that the Fisher information matrix (slide 24 of our slides at <https://docs.google.com/presentation/d/1v4nKhoZKMfotg-h5SrhUuNyM7gWcY9ogo58mX19GTYA/edit?usp=sharing>) is positive semi-definite.

Proof:

Let X be a random variable with density (or mass) $p(x; \theta)$ depending on a parameter $\theta \in \Theta \subset \mathbb{R}^d$. Assume the usual regularity conditions that allow interchange of derivative and integral and ensure that all expectations below exist.

Define the **score function** (vector-valued) as

$$s(X; \theta) := \nabla_\theta \log p(X; \theta) \in \mathbb{R}^d.$$

The Fisher information matrix is defined as

$$I(\theta) := \mathbb{E}_\theta[s(X; \theta)s(X; \theta)^\top].$$

To show that $I(\theta)$ is positive semidefinite, let $a \in \mathbb{R}^d$ be any vector. Then

$$\begin{aligned} a^\top I(\theta) a &= a^\top \mathbb{E}_\theta[s(X; \theta)s(X; \theta)^\top] a \\ &= \mathbb{E}_\theta[a^\top s(X; \theta)s(X; \theta)^\top a] \\ &= \mathbb{E}_\theta[(a^\top s(X; \theta))^2]. \end{aligned}$$

Since $(a^\top s(X; \theta))^2 \geq 0$, its expectation is also nonnegative. Therefore,

$$a^\top I(\theta) a \geq 0, \quad \forall a \in \mathbb{R}^d.$$

Hence $I(\theta)$ is **positive semidefinite**.

3. In class, we have studied the chi-square distribution. In slides 113 and 114 of “Families of Random Variables”, we made an argument that the sample variance obeys a chi-square distribution with $n - 1$ degrees of freedom where n is the number of samples $\{X_i\}_{i=1}^n$. The derivation for the chi-square distribution in those slides made some assumptions, and proving them was postponed as that proof required multivariate statistics. We will complete that gap here. Let $X_1, \dots, X_n$ be iid random variables each having a normal distribution with mean μ and variance σ^2 . Then we define $\bar{X} := \sum_{i=1}^n X_i/n$ and sample variance $S^2 := \sum_{i=1}^n (X_i - \bar{X})^2/(n - 1)$.

Then, we want to prove that $\bar{x}$ and S^2 are independent. We will also prove that $(n-1)S^2/\sigma^2$ is a chi-square random variable with $n-1$ degrees of freedom.

Solution: The solution can be found in Theorem 3.6.1 from the book ‘Introduction to Mathematical Statistics’ by Hogg, Mackean, Craig. This book can be found online at <https://minerva.it.manchester.ac.uk/~saralees/statbook2.pdf>. The equation (3.6.14) requires some more explanation. The two terms on the RHS of (3.6.13) are independent. Hence the MGF of their sum is equal to the product of their individual MGFs. The MGF of the LHS of (3.6.13) is $(1-2t)^{-n/2}$ as V is a chi-square random variable with n degrees of freedom. Hence the MGF of the first term on the RHS of (3.6.13), i.e. the MGF of $(n-1)S^2/\sigma^2$, equals $(1-2t)^{-n/2} \times (1-2t)^{-1/2} = (1-2t)^{-(n-1)/2}$ which is the MGF of a chi-square random variable with $n-1$ degrees of freedom.

4. Derive the Fisher information of the parameter of the exponential distribution.

Solution:

The form parameterized by rate λ The probability density function (pdf) of the exponential distribution is

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0, \lambda > 0.$$

For a single observation x , the log-likelihood is

$$\ell(\lambda) = \log f(x; \lambda) = \log \lambda - \lambda x.$$

The score function is given as follows:

$$s(\lambda) = \frac{\partial}{\partial \lambda} \ell(\lambda) = \frac{1}{\lambda} - x.$$

The Fisher information for one sample is given as follows:

$$I(\lambda) = \mathbb{E}_\lambda[s(\lambda)^2] = \mathbb{E}_\lambda\left[\left(\frac{1}{\lambda} - X\right)^2\right].$$

Since for an exponential distribution $\mathbb{E}[X] = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$,

$$I(\lambda) = \text{Var}(X) = \frac{1}{\lambda^2}.$$

The Fisher information for n independent samples is

$$I_n(\lambda) = n I(\lambda) = \frac{n}{\lambda^2}.$$

We will also do the form of the Fisher information with second derivatives:

$$\ell''(\lambda) = \frac{\partial^2}{\partial \lambda^2} \ell(\lambda) = -\frac{1}{\lambda^2},$$

so that

$$I(\lambda) = -\mathbb{E}_\lambda[\ell''(\lambda)] = -\mathbb{E}_\lambda[-\frac{1}{\lambda^2}] = \frac{1}{\lambda^2}.$$

The form parameterized by the mean $\mu = 1/\lambda$

The pdf can be written as

$$f(x; \mu) = \frac{1}{\mu} \exp\left(-\frac{x}{\mu}\right), \quad x \geq 0, \mu > 0.$$

We have the following log likelihood

$$\ell(\mu) = -\log \mu - \frac{x}{\mu},$$

which yields the following Score function:

$$s(\mu) = \frac{\partial}{\partial \mu} \ell(\mu) = -\frac{1}{\mu} + \frac{x}{\mu^2}.$$

The Fisher information is as follows:

$$I(\mu) = \mathbb{E}_{\mu}[s(\mu)^2] = \mathbb{E}_{\mu} \left[\left(-\frac{1}{\mu} + \frac{X}{\mu^2} \right)^2 \right].$$

Expanding and simplifying using $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \mu^2$:

$$I(\mu) = \frac{\text{Var}(X)}{\mu^4} = \frac{\mu^2}{\mu^4} = \frac{1}{\mu^2}.$$

For n independent samples, we have

$$I_n(\mu) = n I(\mu) = \frac{n}{\mu^2}.$$

Since $\lambda = 1/\mu$ and $\frac{d\lambda}{d\mu} = -\frac{1}{\mu^2}$, the transformation rule gives

$$I(\mu) = I(\lambda) \left(\frac{d\lambda}{d\mu} \right)^2 = \frac{1}{\lambda^2} \left(\frac{1}{\mu^4} \right) = \frac{1}{\mu^2}.$$

Thus the two parameterizations are consistent.